



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

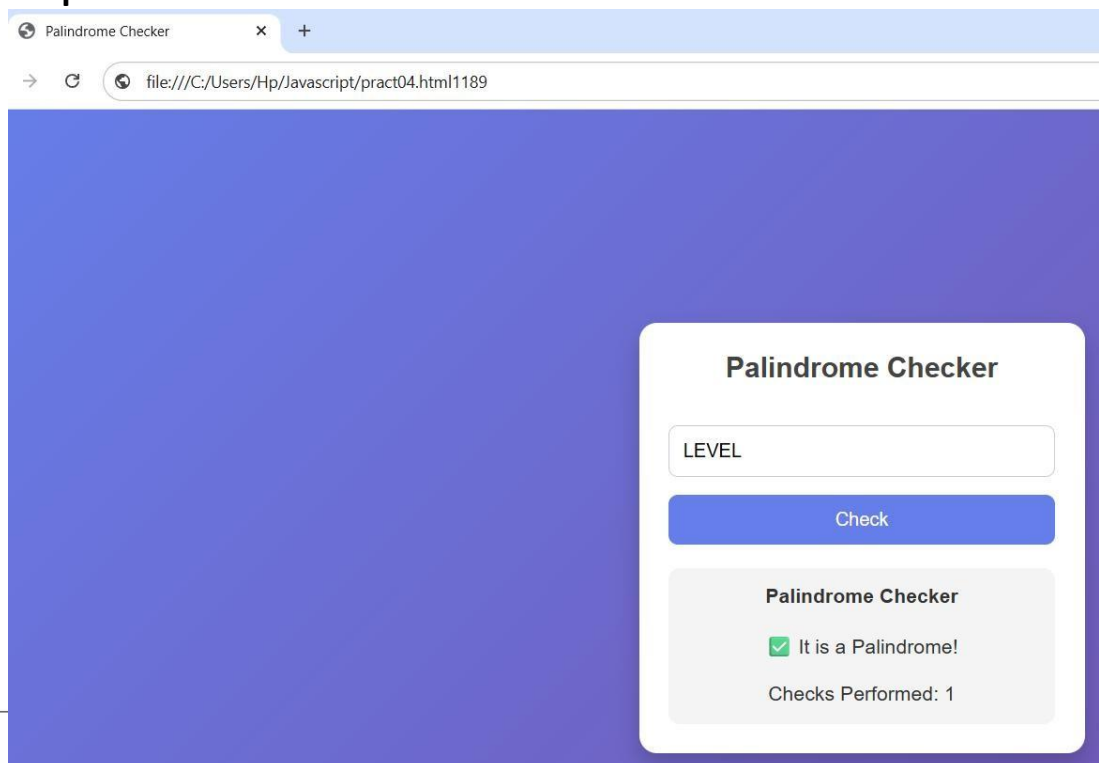
Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 04

- 1. Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.
- 2. Software/Tools Required:** HTML, JavaScript, CSS, VS Code.
- 3. Theory:** This program implements a Palindrome Checker using HTML, CSS, and JavaScript. It allows the user to enter a word or sentence and checks whether it is a palindrome. The program demonstrates different types of functions, including a function declaration, function expression, and arrow function. It also illustrates variable scope by using both global and local variables. A closure is implemented to count the number of palindrome checks performed while preserving the counter value between function calls. Exception handling is achieved using try...catch, which displays an appropriate error message if the input is empty. The webpage dynamically displays the result without refreshing the page.

4. Output:





SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Palindrome Checker

×

+

→ ↺ file:///C:/Users/Hp/Javascript/pract04.html1189

Palindrome Checker

Check

Palindrome Checker

✗ It is NOT a Palindrome.

Checks Performed: 2



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

CASE STUDY 04

Case Study Title: To implement an ATM Card PIN Verification system using JavaScript by demonstrating function declaration, function expression, arrow functions, variable scope, closures, and exception handling.

Code:

```
Welcome Practical03.html pract04.html department.html billing.html index.html
ATM.html > html
2 <html lang="en">
73 <body>

87 <script>
88
89 // Global Scope
90 let storedPIN = "1234";
91
92 // Function Declaration
93 function createCounter(){
94     let attempts = 0;
95
96     // Closure
97     return function(){
98         attempts++;
99         return attempts;
100     };
101 }
102
103 const counter = createCounter();
104
105 // Function Expression
106 const isValidPIN = function(pin){
107     return /^\d{4}$/.test(pin);
108 };
109
110 // Arrow Function
111 const checkPIN = pin => pin === storedPIN;
112
113 // Main Function
114 function verifyPIN(){
115
116     try{
117
118         let pin = document.getElementById("pin").value;
119
120         if(pin=="")
121             throw "Please enter your PIN.";
122
123         if(!isValidPIN(pin))
124             throw "PIN must contain exactly 4 digits.";
125
126         let message;
127
128         if(checkPIN(pin))
129             message = "✅ PIN Verified Successfully!";
130         else
131             message = "❌ Incorrect PIN.";
132
133         document.getElementById("result").innerHTML =
134             message +
135             "<br><br>Verification Attempts: " + counter();
136
137     }
138
139     catch(error){
140
141         document.getElementById("result").innerHTML =
142             "<span style='color:red'>" + error + "</span>";
143
144     }
145 }
```



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Output:

The screenshot shows a web browser window with a single tab titled 'ATM PIN Verification'. The address bar displays the file path: `file:///C:/Users/Hp/Javascript/ATM.html1189Rutva`. The main content area has a dark blue background. On the right side, there is a white card with the title 'ATM PIN Verification'. Inside the card, there is a text input field containing four dots (****). Below the input field is a dark blue button labeled 'Verify PIN'. At the bottom of the card, there is a light gray box containing a red 'X' icon, the text 'Incorrect PIN.', and 'Verification Attempts: 2'.

Result/Conclusion: The Palindrome Checker successfully demonstrates important JavaScript concepts such as function types, variable scope, closures, and exception handling (try...catch). It validates user input, processes the entered text by removing spaces and special characters, and determines whether it is a palindrome. The project provides a user-friendly interface and showcases how JavaScript can be used to build interactive, reliable, and efficient web applications.